

AI-Based Medical Diagnosis Expert System v2.0

An AI-powered expert system that assists users in identifying possible diseases based on symptoms. Built with Prolog for knowledge representation and reasoning, and Python for the graphical user interface.

Features

Core Features

- Disease prediction from **25+ diseases**
- Symptom selection from **50+ symptoms**
- Rule-based inference using Prolog
- Weighted confidence scoring with severity weights
- Detailed explanation of diagnosis results
- Forward and backward chaining support

Advanced Features

- **Patient Profile Management** - Save patient information (name, age, gender, blood type)
- **Diagnosis History** - Track all past diagnoses with timestamps
- **Statistics Dashboard** - Visual charts showing disease distribution and symptom frequency
- **Disease Encyclopedia** - Browse all diseases with descriptions, categories, and recommendations
- **Emergency Flags** - Critical conditions are highlighted for immediate attention
- **Symptom Search & Filter** - Search symptoms by name or filter by body system category
- **Dark/Light Theme** - Toggle between light and dark modes
- **Export Reports** - Export diagnosis results and history to text files
- **Disease Categories** - Diseases organized by type (viral, bacterial, respiratory, etc.)
- **Body System Organization** - Symptoms grouped by body system for easier selection

Prerequisites

- Python 3.8+
- [SWI-Prolog](https://www.swi-prolog.org/) (must be installed and in system PATH)

Installation

1. Install SWI-Prolog from <https://www.swi-prolog.org/>
2. Ensure SWI-Prolog is added to your system PATH
3. Install Python dependencies:

```
pip install -r requirements.txt
```

Usage

```
python gui.py
```

1. Set up your patient profile (optional)
2. Select symptoms from the organized checklist
3. Click **Diagnose** to see possible diseases
4. View confidence scores, categories, and explanations
5. Export reports or view history and statistics

Project Structure

```
Medical_Expert_System/
├── gui.py                # Enhanced Tkinter GUI with tabs, themes, charts
├── controller.py         # Python-Prolog bridge (PySwip) with search/filter
├── medical_kb.pl         # Prolog knowledge base (25 diseases, 50+ symptoms)
├── patient_profile.py    # Patient profile and history management
├── report_exporter.py    # Report export to TXT format
├── requirements.txt      # Python dependencies
├── data/                # Auto-created: stores patient data and history
├── assets/              # Images and resources
└── screenshots/         # Application screenshots
```

Knowledge Base

Diseases (25+)

Category	Diseases
Viral	Flu, Cold, COVID-19, Dengue, Chickenpox, Measles, Hepatitis A
Bacterial	Typhoid, Tuberculosis, UTI
Respiratory	Asthma, Pneumonia, Bronchitis
Neurological	Migraine, Insomnia
Metabolic	Diabetes
Cardiovascular	Hypertension
Other	Malaria, Anemia, Allergy, Sinusitis, Gastroenteritis, Heat Stroke, Depression, Arthritis

Symptoms (50+)

Organized by body system: General, Respiratory, Neurological, Gastrointestinal, Musculoskeletal, Skin, Cardiovascular, Urinary, Psychological, ENT

Reasoning

- **Backward Chaining:** Prolog's default goal-driven reasoning
- **Forward Chaining:** Data-driven inference from symptoms to diseases
- **Weighted Confidence:** Severity-weighted symptom matching for more accurate scoring
- **Emergency Detection:** Flags life-threatening conditions automatically

Technologies

- Python 3
- SWI-Prolog
- PySwip (Python-Prolog bridge)
- Tkinter (GUI)
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Disclaimer

This system is for **educational and demonstration purposes only**. It is NOT a substitute for professional medical diagnosis. Always consult a qualified healthcare provider for medical advice.